

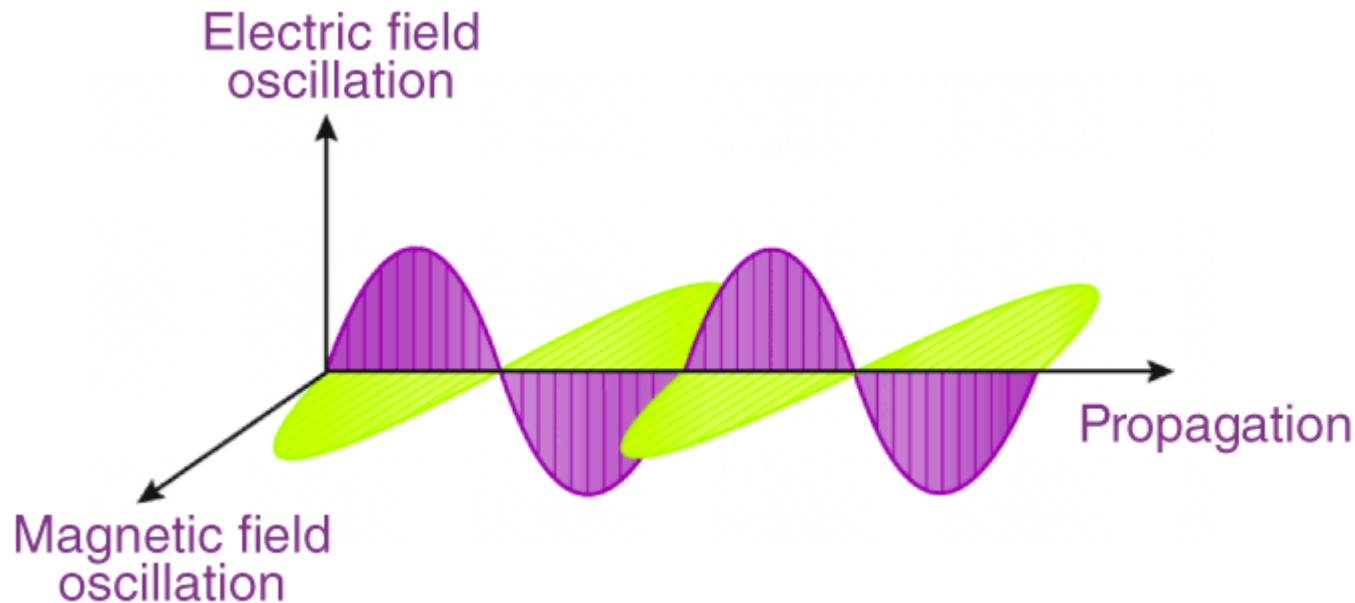
Interaction of radiation with matter

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Electromagnetic waves



- Electromagnetic waves were first postulated by James Clerk Maxwell and subsequently confirmed by Heinrich Hertz.
- In this chapter, we will explore the definition and formation of electromagnetic waves along with the graphical and mathematical representations of electromagnetic waves in detail.
- Electromagnetic radiations are composed of electromagnetic waves that are produced when an electric field comes in contact with the magnetic field. It can also be said that electromagnetic waves are the composition of oscillating electric and magnetic fields travelling through free space with the speed of light c .
- Electromagnetic waves transfer energy through space.

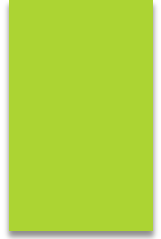


- Electromagnetic waves are shown by a **sinusoidal graph**. It consists of **time-varying electric and magnetic fields** which are **perpendicular to each other** and are also **perpendicular to the direction of propagation of waves**.
- Electromagnetic waves are transverse in nature.
- The highest point of the wave is known as the crest while the lowest point is known as a trough.
- In vacuum, the waves travel at a constant velocity of $3 \times 10^8 \text{ m.s}^{-1}$.

How Are Electromagnetic Waves Formed?

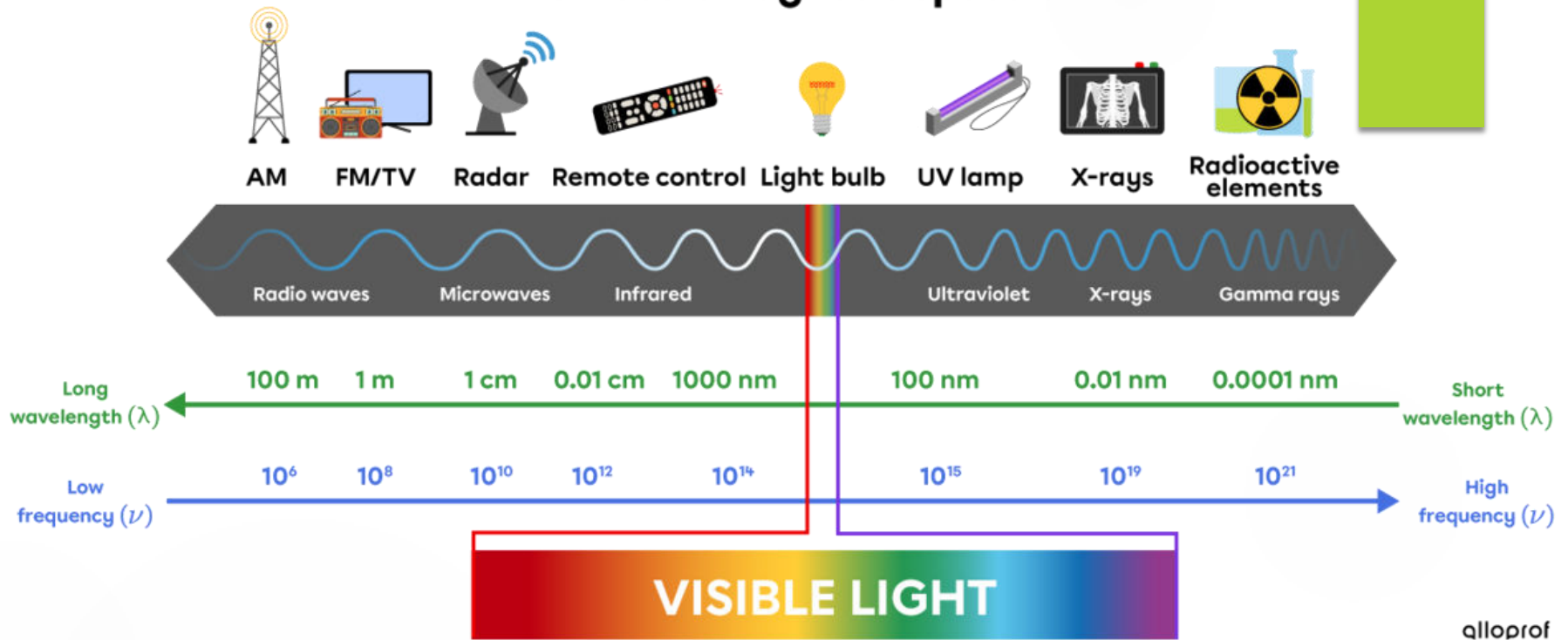
- Generally, an **electric field** is produced by a charged particle. A force is exerted by this electric field on other charged particles. Positive charges accelerate in the direction of the field and negative charges accelerate in a direction opposite to the direction of the field.
- The **Magnetic field** is produced by a moving charged particle. A force is exerted by this magnetic field on other moving particles. The force on these charges is always perpendicular to the direction of their velocity and therefore only changes the direction of the velocity, not the speed.
- So, the electromagnetic field is produced by an accelerating charged particle. An accelerating charged particle is when the charged particle oscillates about an equilibrium position .

Electromagnetic Spectrum



- The EM spectrum spans the entire range of electromagnetic radiation from the lowest frequency/longest wavelength to the highest frequency/shortest wavelength. Various segments of the spectrum are useful in exploring the Universe, and enable us to "see" into the universe beyond what we are able to see from visible light.
- If the frequency of oscillation of the charged particle is f , then it produces an electromagnetic wave with frequency f . The wavelength λ of this wave is given by $\lambda = c/f$.
- Electromagnetic waves are classified according to their frequency f or according to their wavelength $\lambda = c/f$.

The Electromagnetic Spectrum



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- Visible light, radio waves and X-rays are examples of **electromagnetic radiation** (EMR).
- Waves with a long wavelength spread well beyond their surroundings. **Radio** waves are used to transmit wireless signals to TVs, radios and mobile phones through the air.
- **Microwaves** are less widespread and are used where the transmitter and the receiver are closer to each other. In this spectrum you will, for example, find radar and microwave ovens..

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- Physical bodies with temperatures around room temperature and above will emit radiation in the **infrared** area. This is radiation in the form of heat. Different cameras and surveillance equipment such as motion detectors can react to infrared radiation.
 - **Visible light** has a whole spectrum of colours. From the colour **red** with the **longest wavelength** (and thus least energy) to the colour **violet** which has the **shortest wavelength** (and the highest energy):
 - **X-rays** are energy rich radiation and can penetrate soft tissue. This type of radiation does not penetrate bones, and is therefore very suitable for photographing the skeleton of humans and animals.
 - **Gamma** radiation is very energy rich and penetrates most solid materials. Gamma rays are ionizing radiation and can therefore destroy molecules and connections in atoms and cause damage to tissues.

Exercise: A ray, emitted from the sun, is shining through your kitchen window into a prism. The prism then casts a rainbow on the windowsill. You just so happen to have a hand-held radiometer handy; closing your eyes you place the instrument on a specific color of the rainbow. You then open your eyes to see that the radiometer measured the energy from that color at 3.0×10^{-19} joules. Given Plank's constant of 6.6256×10^{-34} joules/sec, what possible color did the radiometer measure? How do we know this?

Solution:

Energy change = 3.0×10^{-19} joules

Plank's constant = 6.6256×10^{-34} joules/sec

$$E = h \cdot f$$

$$(3.0 \times 10^{-19} \text{ joules}) / 6.6256 \times 10^{-34} \text{ joules/sec} = f$$

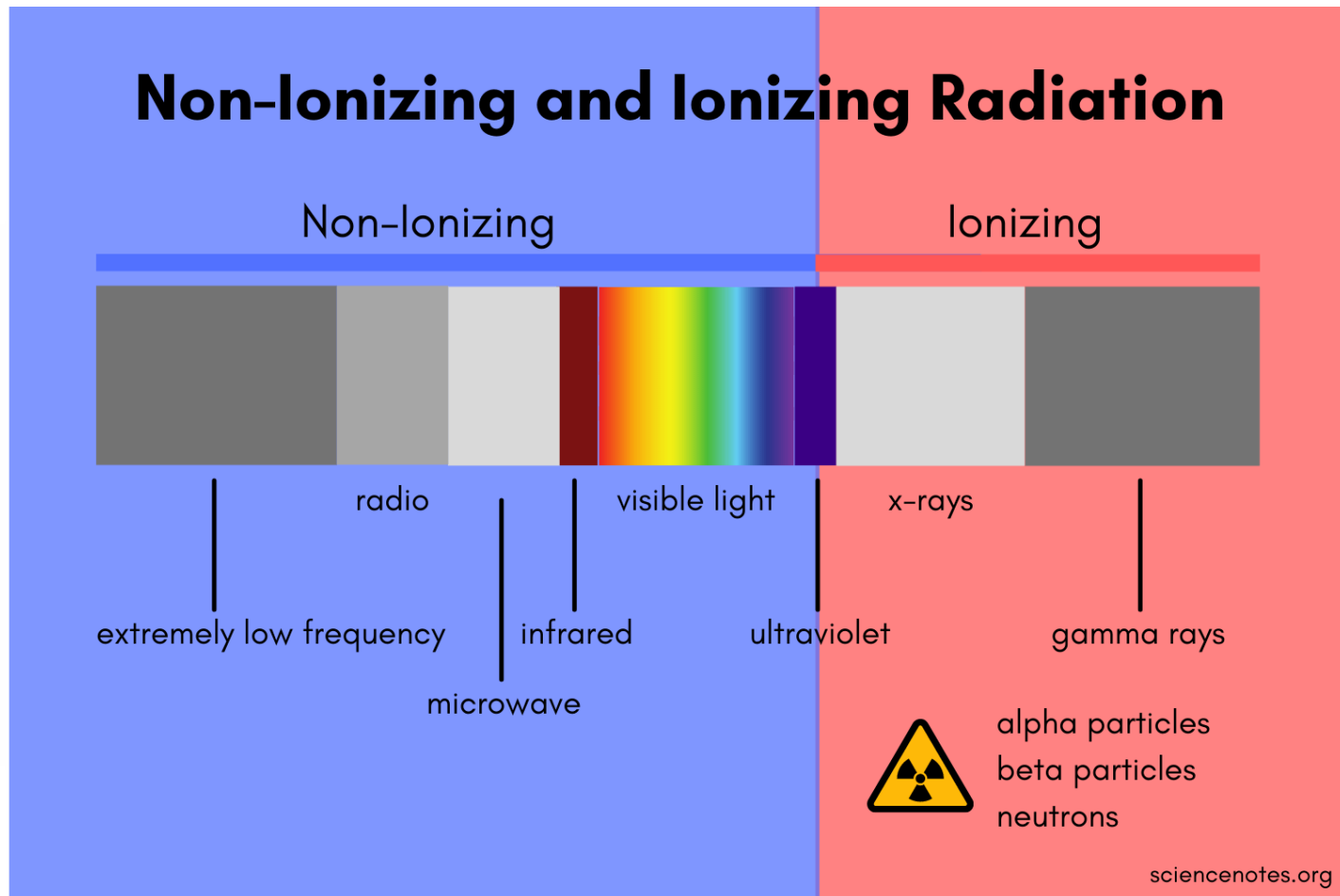
$$= 4.5 \times 10^{14} \text{ sec}^{-1}$$

*This falls within the given visible spectrum frequencies. Being a low frequency on the visible spectrum, close to the frequencies emitted by red colors, **one could assume that you measured a color close to red; though orange and yellow cannot be out of the question as the given frequencies are a bit vague.***

Types of Radiation

As we know, radiation is energy emitted by a source and then travels through a medium, such as air, until it is absorbed by matter. Based on the energy of the radiated particles, they are classified as follows:

1. Ionizing radiation
2. Non-ionizing radiation





Ionizing radiation is caused due to unstable atoms that either have excess energy, mass, or both. To return to a stable state, they have to release the extra mass or energy in the form of radiation.

Different types of ionizing radiation:

1. Alpha particles
2. Beta particles
3. Gamma rays
4. X-rays

Uses of Alpha Particles

- They are used as smoke detectors. When the smoke enters the detector, the number of alpha particles reduces and gets detected by the alarm. Americium-241 is an example of a smoke detector



Uses of Beta Particles

- Strontium-90 is used for the treatment of bone cancer
- For emergency lighting, tritium is used.

Uses of Gamma Rays

- Cobalt-60 is used for sterilisation of medical equipment and pasteurisation of food.
- Caesium-137 is used to measure soil moisture density and control the flow of liquids in industries.

Uses of X-rays

- X-rays are used for crystallography and fluoroscopy to control the quality of materials.
- X-rays are used to detect tumours and infections.



Non – Ionizing Radiation

Non-ionizing radiation is the term given to radiation that has insufficient energy to cause ionization. These kinds of radiation contradict ionizing radiation like x-rays, gamma rays, and alpha particles, which are on the other end of the spectrum and are unstable and reactive. Non-ionizing radiation can produce heat, which is how food is cooked in a microwave oven. Humans and other organisms can see some types of non-ionizing radiation, such as visible light and infrared light.

Following are the types of non-ionizing radiation:

Ultraviolet radiation

Visible light

Infrared

Microwave

Radio waves

Uses of Ultraviolet Rays

- UV radiation is widely used in industrial processes and in medical and dental practices for various purposes, such as killing bacteria, creating fluorescent effects, curing inks and resins, phototherapy and suntanning.

Uses of Visible Light

- Visible light used in fiber optic communications, photography, and electronic devices.

Uses of infrared waves

- Thermography is a method to determine the relative temperature of the object, and for this, infrared waves are used.
- Infrared waves are invisible but can be sensed as they conduct heat. Because of this property, infrared waves are used in physical therapy for heating purposes.

Uses of Microwaves

- They are used in communications, radio astronomy, remote sensing, radar, and of course, owing to their heating application, they are used in cooking as well.

Ionising vs Non Ionising Radiation

More Information Online WWW.DIFFERENCEBETWEEN.COM

DEFINITION

Ionising Radiation
Ionising radiation is high energy radiation, which can ionize an atom upon colliding with it to emit more radiation.

Non Ionising Radiation
Non ionising radiations are low energy radiation in which it does not emit particles from other materials.

ENERGY

High energy

Low energy

EMISSIONS

Can emit electrons or other particles from atoms when they collide.

Cannot emit particles from an atom; can only excite electrons from a lower level to a higher level upon encountering

EXAMPLES

Gamma, X-ray

UV, visible, IR, microwave and radio waves

Particle properties of wave

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Introduction

Classical mechanics or Newtonian mechanics deals with **macroscopic** systems-like big bodies, systems of huge number of atoms or molecules that are directly observable or observable with instruments like microscope. Classical mechanics concerns the motion of a particle under the influence of applied forces. This includes measurements of particle's position, mass, velocity, acceleration etc. Classical mechanics predicts for observable magnitudes (say: $v = u + at$, where u , a , t are given) agree with measured values.

However, the particles which are not observable, can we apply laws of classical mechanics for them? No, classic concepts cannot be applied to observe the behavior of atoms, molecules, electrons, protons, nuclei, etc. These are called **microscopic particles**. At the end of the 19th century, with the discovery of electrons, x-rays and radioactivity, the possibility arose to study the behavior of individual atoms or molecules. But the results obtained from the studies could not be explained by the classical mechanics. Then a new theoretical branch of physics, called **Quantum mechanics** was developed by Erwin Schrödinger, Warner Heisenberg, P. M. Dirac and others.

Inadequacy of classical physics and concept of Quantum

- i) Consider the case of an electron moving round the nucleus. Classical mechanics tells that accelerated charged particle loses energy in the form of electromagnetic waves. Therefore, its velocity should decrease continuously. The electron should come closer to the nucleus until it collapses. Actually, the atom does not collapse. So, classical mechanics fails to explain the stability of atom.

